ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/386401880

Brief report: Exploring the icing the kicker strategy in the NFL

Article *in* Journal of Applied Sport Psychology · December 2024

DOI: 10.1080/10413200.2024.2437166

CITATIONS
0

READS
68

3 authors, including:

Nadav Goldschmied
University of San Diego
40 PUBLICATIONS **576** CITATIONS

SEE PROFILE

Mike Raphaeli
Nanning Vocational and Technical University
7 PUBLICATIONS **37** CITATIONS

SEE PROFILE

All content following this page was uploaded by Nadav Goldschmied on 08 December 2024.

The user has requested enhancement of the downloaded file.

Journal of Applied Sport Psychology

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/uasp20

Brief report: Exploring the icing the kicker strategy in the NFL

Nadav Goldschmied, Tyler Ratkovich & Mike Raphaeli

To cite this article: Nadav Goldschmied, Tyler Ratkovich & Mike Raphaeli (03 Dec 2024): Brief report: Exploring the icing the kicker strategy in the NFL, Journal of Applied Sport Psychology, DOI: 10.1080/10413200.2024.2437166

To link to this article: https://doi.org/10.1080/10413200.2024.2437166

View supplementary material

Published online: 03 Dec 2024.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=uasp20

JOURNAL OF APPLIED SPORT PSYCHOLOGY
https://doi.org/10.1080/10413200.2024.2437166

Routledge
Taylor & Francis Group

NOTE

Brief report: Exploring the icing the kicker strategy in the NFL

Nadav Goldschmied[a], Tyler Ratkovich[a], and Mike Raphaeli[b]

[a]University of San Diego College of Arts and Sciences; [b]Nanning Vocational and Technical University

ABSTRACT

Icing is a strategy used in American football during the last moments of game when a coach calls for a time-out to allow an opposing kicker to supposedly ruminate before a decisive field-kick. Using archival data of pressure kicks from twenty-five National Football League seasons (1999–2024), a generalized estimating equation (GEE) model accounting for nested data was applied. It was found that icing was successful in reducing scoring while other factors such as location (home vs. away), physical environment (indoors vs. outdoors) or season type (regular vs. playoffs) were not associated with the kicks' success. In a secondary analysis it was demonstrated with another limited set of nullified kicks that the second attempt in close time proximity did not yield a significant deterioration in performance thus presently failing to lend support to the "twice taxed" hypothesis (Goldschmied et al., 2010) of why icing works. Theoretical and applied implications are discussed.

Lay Summary: Attempts at icing or the calling of time-outs by opposing coaches against kickers in the National Football League before crucial kicks were demonstrated to be successful (1999–2024 seasons). Other factors such as home vs. away, indoors vs. outdoors or regular vs. playoffs were not associated with the kicks' success.

IMPLICATIONS FOR PRACTICE

- Opposing coach should preserve and use time-outs to ice the kicker before crucial kicks.
- Kickers should practice kicking with icing manipulations in training.

ARTICLE HISTORY
Received 8 August 2024
Revised 17 November 2024
Accepted 22 November 2024

The fate of an NFL (National Football League) game is often placed on the shoulders (or more precisely legs) of the kicker (Hsu et al., 2019) with an increasing frequency (Associated Press, 2023). This specialized undertaking is a focus of intense study in an effort to isolate factors associated with its successful execution (Goldschmied et al., 2010; Gonsalves, 2013; Morrison & Kalwani, 1993). To score a field goal (FG), the team in possession of the ball must kick the ball through the opposing team goal's uprights over a crossbar that is 10′ high (3.05 m) and 18'6" long (5.64 m). A successful field goal

CONTACT Nadav Goldschmied ngoldschmied@sandiego.edu Psychological Sciences, University of San Diego College of Arts and Sciences, San Diego, CA, USA.

Supplemental data for this article can be accessed online at https://doi.org/10.1080/10413200.2024.2437166.

© 2024 Association for Applied Sport Psychology

in the NFL awards three points (as opposed to a touchdown, or the advancing the ball into the designated goal area which is worth six points).

While the success of the FG is mostly dependent on the distance of the kick (Goldschmied et al., 2010), some psychological based correlates, specifically "icing the kicker," were found to be detrimental to the success of this effort. Icing is an act in which the opposing coach extends time by calling a time-out (TO) just before an opponent player is about to execute a routinely practiced skill when the game is on the line (Goldschmied et al., 2023). In football, this maneuver is exercised to presumably induce rumination and self-doubt in the rival kicker before the FG attempt as other defensive maneuvers to block the kick are rarely successful (e.g., rushing the kicker).

Berry and Wood (2004) showed that kickers' performance declined in the NFL after being iced before pressure kicks. These researchers classified a pressure kick as one that would create a lead or tie the game for the kicking team within three or less minutes remaining in the game (or overtime) for the 2002 and 2003 NFL regular seasons and playoffs. Kickers were successful 73% of the time when kicking when not iced (101 successful kicks out of 139 attempts), while managing only 67% of the kicks after having been iced (24 successful kicks out of 38 attempts). However, their conclusions were tempered by the small sample size.

Subsequently, Goldschmied et al. (2010) studied pressure kicks from six consecutive NFL seasons (2002–2008). Once again, it was found that icing was successful as in the 110 instances where kickers were iced, they were able to score only 73 of the kicks (66.4%), while among the 163 not iced kicks, 131 kicks were converted (80.4%). In a follow-up analysis, it was shown that if the coaches of the kicking team called a TO before the pressure kick with apparently the intention to bring order and reduce confusion in their ranks (as opposed to a TO called by the opposing team for icing), kickers no longer faltered. Since the kickers were still subjected to the same time extension when their own coach initiated the TO, the researchers surmised that a swelling self-doubt and negative rumination may not be at the root of the effect. They instead postulated that mental and physical over-exertion led to reduced performance as the TO was initiated unexpectedly by the opposing coach which they termed as the "twice-taxed" hypothesis.

Clark et al. (2013) included all FG attempts from the 2000–2011 NFL seasons and found that icing did not affect the outcome of the kick. Only environmental factors such as distance and temperature were negatively correlated with its success. Lastly and most comprehensively, Hsu et al. (2019) studied all NFL FG attempts from the 2000 to 2017 seasons. The researchers first found that any TO (called by either coach) was detrimental to kicking performance. Classifying "icing" as a TO called by the opposing coach had some negative effect as well, but not significant.

The two latter and more recent studies of Clark et al. (2013) and Hsu et al. (2019) are more comprehensive in scope and were thus enhanced in their statistical power since they did not limit their exploration to end of game pressure kicks. However, in doing so, they disregarded the many studies that found that end of game performance when the score is close presents greater psychological burden that is likely to hurt performance (for example, Bar-Eli & Tractinsky, 2000; Goldschmied et al., 2023; Toma, 2017).

In the current investigation, since past research on icing the kicker was conflicting in its results, we endeavored to study if indeed FG kicking was different in its execution and success in close end of games situations than otherwise. We then introduce a regression model with a substantially expanded sample size over past research and target exclusively end of game pressure situations.

Our prediction model includes a modified and concise conceptualization of the environmental predictor variables. Past research utilized temperature, wind speed and surface type separately (Clark et al., 2013; Hsu et al., 2019) in the attempt to provide a nuanced approach. However, the environmental data are tabulated at the beginning of the game and not at the specific time of the kick and wind direction is not reported. A second problem to arise from the many environmental factors utilized is multicollinearity or the inflation of the variance or standard error of coefficient estimates due to great correlation among the predictor variables (Mela & Kopalle, 2002). We thus opted to consolidate all environmental circumstances in our model into a single variable, namely whether the kick was performed indoors (i.e., protected from the elements, little to no variability in environmental variables) or outdoors (some variability in environmental variables). This inclusive classification should also increase the statistical power (Clark et al., 2013).

Lastly, in addition to including all NFL FG attempts two minutes toward the end of regulation or overtime when the game was close, we identified another set of FG attempts in which a kick was undertaken but then due to an infringement committed during the play, it had to be repeated. These rare kicking instances which occur at any point in a game (and not only in close, end-of-game situations) could shed light whether the idea of the kicker being mentally and physically "taxed twice" (Goldschmied et al., 2010) could explain the rationale for why icing is a successful strategy.

Methods

We obtained data from 25 NFL seasons (1999–2024) through https://github.com/nflverse.

GitHub is a platform that allows users to create, store, manage and share code. All data are available at https://osf.io/bxk5p/. For the descriptive statistics of factors studied see Table 1.

A pressure kick in this study was defined as one that was performed within two minutes or less of the end of regulation time when the kicking team was behind by three points or less (ties included) and the success or failure of the kick would determine whether a team could tie the game or place them in the lead to win the game. These four game score situations (-3, −2, −1, tied) within regulation were classified as "Highest pressure" condition in Clark et al. (2013) 6-category Pressure scheme. We also included all kicks during overtime, which even if successful, give the opposing team the opportunity to score subsequently and possibly win the game (each team gets two timeouts during this period). This game condition was classified as "Higher stress" category in Clark et al. (2013) 6-category Pressure scheme. Jackson (2003) used a similar scheme in the analysis of rugby World-cup goal kicking success.

Table 1. Comparison of descriptive statistics between iced, no timeout and timeout by own team pre- pressure kicks.

Variable	Iced kicks (n = 429)	No timeout (n = 550)	Timeout by own team (n = 311)
Distance	40.88 +/- 10.22	40.65 +/- 10.85	37.65 +/- 10.48
Location:			
Home kicking	227 (52.9%)	287 (52.1%)	156 (50.2%)
Away kicking	202 (47.1%)	264 (47.9%)	155 (49.8%)
Physical environment			
Indoors	124 (28.9%)	161 (29.2%)	95 (30.5%
Outdoors	305 (71.1%)	390 (70.8%)	216 (69.5%)
Season			
Regular	397 (92.5%)	531 (96.4%)	298 (95.8%)
Playoffs	32 (7.5%)	20 (3.6%)	13 (4.2%)
Conversion ratio:			
Yes	318 (74.1%)	434 (78.8%)	253 (81.4%)
No	111 (25.9%)	116 (21.1%)	58 (18.6%)

Note: Data are shown as mean +/- standard deviation for the continuous variables and as absolute (relative) frequencies for the categorical variables.

A second set of kicks (also available at https://osf.io/bxk5p/) were intended to explore the "twice taxed" hypothesis whereupon in a second kick upon the nullification of the first one, the kicker is seemingly experiencing resource depletion (Goldschmied et al., 2010) which brings about deterioration in performance. We identified 106 such kicks during the respective 25 seasons (excluding "triplets," blocked and critical conditions kicks). It is important to note that these successive kicking efforts are not exact replications of each other as there is a slight difference in distance between the first and second attempt. When the kicking team is successful but commits an infraction during the play it is then penalized and has to kick again from further away. Alternatively, if the kick was unsuccessful but the defensive team commits an infraction then the subsequent attempt is undertaken from a closer distance to the goal.

Results

First, we compiled the success rate of FG kicks in end of game pressure situations as opposed to ones performed during the rest of the game. A Paired samples t-test (by season) based on cumulative seasonal success (makes over attempts * 100) indicated a significant deterioration in performance under pressure ($M = 77.76\%$; $SD = 4.74$) than during regulation ($M = 84.32\%$; $SD = 2.69$); [$t(24) = 6.425$, $p < .001$; $d = 1.285$, representing a large effect size, according to Cohen (1992)]. Moreover, to distinguish the two game situations further, we conducted a correlation analysis to assess improvement over time (seasons). Kicking success during non-critical game situations showed consistent improvement over time, $r(25) = .82$, $p < .001$ (since the 2002 season it never dipped below 80%), however, the same trend was not observed with kicking under pressure, $r(25) = -.013$, $p = .95$.

For our main analysis, we employed a generalized estimating equation (GEE) to account for the correlation between multiple observations in the dataset. Since NFL kickers kick many pressure kicks, a traditional regression method could not be used for the risk of violating the statistical assumption of the independence of observations (Ghisletta & Spini, 2004).

Table 2. Generalized estimating equation model for successful conversion of pressure kicks (binary logistic).

Parameter	B	S.E.	95% Wald CI	Sig.
I. Model 1:				
Timeout pre-kick[icing/no/by own]	.162	.07	.024–.30	.021*
Kick distance	.102	.008	.086–.118	.000*
Arena [indoor/outdoor]	–.322	.188	–.692–.046	.085
Season [regular/playoff]	–.007	.409	–.809–.795	.99
Location [home/away]	.08	.154	–.222–.381	.605
Intercept	– 5.65	.488	–6.61–4.69	.000*
II. Model 2:				
Timeout [icing/no + by own]	–.274	.139	–.547–.002	.049*

Note: The degrees of freedom (*df*) for all the variables is 1. Icing is timeout taken by rival team before the kick. Model 2 has all the same variables as Model 1 except icing, but the confounders are not reported for the sake of brevity.

We identified overall 1,352 pressure kicks in line with the criteria, of which 1005 were successful and 285 were missed. We eliminated from our analysis 42 kicks which were blocked.

Table 2 shows the results of the binary logistic regression (no risk for multicollinearity was observed) which included 935 games. This sample included 128 unique kickers with 108 of them kicking more than once. The results support earlier findings (Goldschmied et al., 2010) that icing undermines the success of an NFL pressure kick. This was shown when we treated TO occurrence before the kick as ordinal with three distinct levels (opponent, absence, own) or as a dichotomous one distinguishing icing from the two other conditions clustered together (model II). We failed to detect significant associations with any of the predictor variables except for the distance of the kick.

We next assessed a subsample of successive kicks due to nullification in "non-critical" game situations. The success rate for the 106 second after-nullification kicks was 78.3%. A chi-square test of goodness-of-fit was performed to determine whether this subset of kicks had lesser success than that of the population of non-critical kicks ($M = 84.32$). While in the direction to support the "double-taxed" hypothesis, the test failed to reach significance, $\chi 2$ (1, N = 106) = 2.52, p = .112, Cohen's W = .15 (small effect size, Cohen, [1992]).

Discussion

In the current research of icing effects in NFL play, the most extensive study of yet, we found that this adversarial strategic ploy did decrease the chances of the kicker to score. The effect took hold whether we included the act of icing against both a no TO situation before the kick as well as a TO taken by the kicker's own team. This reduced rate of success following icing was also evident when we kept the three game situations distinct in our analysis. However, studying the success rate, we observe that the effect size is diminished relative to past research. Goldschmied et al. (2010) tabulated 66.4% success for iced kicks and 80.4% for all other kicks. In the current exploration, kick success when iced was 74.1% as opposed to 79.8% otherwise. A significant improvement in the ability to deal with iced game situation was thus observed while the base rate of success overall was about the same. Some of the observed diminished effect is due to the more restrictive parameters the current investigation utilized as we a-priori eliminated blocked kicks attributing then the unsuccessful outcome on teammates rather than kicker.

Studying icing under the prism of choking under pressure, two broad theoretical models usually are proposed to account for the deterioration (Mesagno & Beckmann, 2017): excessive and undermining self-focus or misdirected attention toward non-task relevant cues (i.e., distraction). The most common belief behind the undermining icing mechanism is that the prolonged time prior to the kick, allows for the kicker's contemplation of the consequences of failure and excessive rumination subsequently (Kröhler & Berti, 2019). Icing may also be disruptive as it either interrupts a pre-performance routine that begins from the moment the kicker is summoned onto the field or it creates a state of uncertainty as to when a pre-performance routine should commence and how long it should be practiced relative to the kicking effort. Gröpel and Mesagno (2019) conducted a systematic review of choking interventions in sports and incorporating the unique conditions of kicking in football, the act of icing may be countered several ways. Initially, instructing kickers to utilize a quiet eye technique placing their final visual fixation always toward a relevant target (i.e., goal posts) prior to the initiation of a movement (Vickers, 2007) has shown promising results. This relatively short and simple act could be easily practiced under pressure icing conditions. Empirical support was also found for the left-hand contractions approach whereupon the exercise activates the visuospatial processes of the right hemisphere essential for motor output and supposedly, in turn, suppresses the analytical processes of the left hemisphere linked to the control of skill execution (again, seemingly a simple practice routine to counter icing). Lastly, simulating icing in practice especially under self-focus ((Beilock & Carr, 2001) and anxiety conditions (Bell et al., 2013) should transfer positively to game conditions. Future research should survey which techniques, if any, are practiced with NFL kickers.

Goldschmied et al. (2010) offered a different icing focus whereby kickers who are iced indeed face uncertainty but placed the emphasis on having to prepare and sometimes kick twice as they were being "doubly-taxed" physically and mentally (while kickers whose own coach is calling the TO, know it is coming in advance and avoid the need to prepare twice). Studying nullified kicks that had to be kicked again in very close time proximity throughout the game (and not in game deciding situations) failed to yield the statistical threshold of support for this idea as the second kick performance did not markedly deteriorate beyond the level of success usually experienced by NFL kicks in non-critical situations. However, we did observe a general reduced performance upon the second attempt.

Going beyond the archival study practiced here, future research should also employ observational as well as self-report efforts to disentangle the mechanism behind the debilitating effects of icing. The current extensive undertaking has substantiated the icing effect in the NFL and explored a possible path to explain the association. While the "doubly-taxed" hypothesis we offered failed to gain empirical support, it is worth noting that our sample size of nullified kicks was relatively small, the successive kicks were attempted from varying distances and pertain to "non-critical" kicks but were applied to end-of-game pressure situations to simulate the icing strategy. Based on these limitations, we think that this hypothesis merits further expanded scrutiny.

Goldschmied et al. (2010) recommended that an opposing coach should preserve and use a TO under the conditions of time and score defined in the current study to ice the kicker. Theoretically, the next meaningful step in this undertaking would be to develop a working model to understand and explain this phenomenon fully.

Acknowledgments

We would also like to thank Emma Rohrer, Nicole Sevcikova, Elliot Gorsuch, Nicole Hunter, Eric Miller and Professor Otten for data collection assistance.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The dataset generated during the current study is available at All data are available at https://osf.io/bxk5p/ or from the corresponding author on reasonable request.

References

Associated Press. (2023). *5 NFL games end on field goals as time expires, most ever in one day*. https://apnews.com/article/nfl-5-games-end-with-field-goals-record-5664cbfc97db75f4358544dca7b9ae81

Bar-Eli, M., & Tractinsky, N. (2000). Criticality of game situations and decision making in basketball: an application of performance crisis perspective. *Psychology of Sport and Exercise*, *1*(1), 27–39. https://doi.org/10.1016/S1469-0292(00)00005-4

Beilock, S. L., & Carr, T. H. (2001). On the fragility of skilled performance: What governs choking under pressure? *Journal of Experimental Psychology. General*, *130*(4), 701–725.

Bell, J. J., Hardy, L., & Beattie, S. (2013). Enhancing mental toughness and performance under pressure in elite young cricketers: A 2-year longitudinal intervention. *Sport, Exercise, and Performance Psychology*, *2*(4), 281–297. https://doi.org/10.1037/a0033129

Berry, S. M., & Wood, C. (2004). A statistician reads the sports pages: The cold-foot effect. *Chance*, *17*(4), 47–51. https://doi.org/10.1080/09332480.2004.10554926

Clark, T., Johnson, A., & Stimpson, A. (2013). *Going for three: Predicting the likelihood of field goal success with logistic regression* [Paper presentation]. The 7th Annual MIT Sloan Sports Analytics Conference, March.

Cohen, J. (1992). A power primer. *Psychological Bulletin*, *112*(1), 155–159. https://doi.org/10.1037/0033-2909.112.1.155

Ghisletta, P., & Spini, D. (2004). An introduction to generalized estimating equations and an application to assess selectivity effects in a longitudinal study on very old individuals. *Journal of Educational and Behavioral Statistics*, *29*(4), 421–437. https://doi.org/10.3102/10769986029004421

Goldschmied, N., Nankin, M., & Cafri, G. (2010). Pressure kicks in the NFL: An archival exploration into the deployment of time-outs and other environmental correlates. *The Sport Psychologist*, *24*(3), 300–312. https://doi.org/10.1123/tsp.24.3.300

Goldschmied, N., Raphaeli, M., & Morgulev, E. (2023). "Icing the shooter" in basketball: The unintended consequences of time-out management when the game is on the line. *Psychology of Sport and Exercise*, *68*, 102440. https://doi.org/10.1016/j.psychsport.2023.102440

Gonsalves, R. (2013). *Placekicking in the NFL: A History and Analysis*. McFarland.

Gröpel, P., & Mesagno, C. (2019). Choking interventions in sports: A systematic review. *International Review of Sport and Exercise Psychology*, *12*(1), 176–201. https://doi.org/10.1080/1750984X.2017.1408134

Hsu, N. W., Liu, K. S., & Chang, S. C. (2019). Choking under the pressure of competition: A complete statistical investigation of pressure kicks in the NFL, 2000–2017. *PLoS One*, *14*(4), e0214096. https://doi.org/10.1371/journal.pone.0214096

Jackson, R. C. (2003). Pre-performance routine consistency: Temporal analysis of goal kicking in the Rugby Union World Cup. *Journal of Sports Sciences*, *21*(10), 803–814. https://doi.org/10.1080/0264041031000140301

Kröhler, A., & Berti, S. (2019). Taking action or thinking about it? State orientation and rumination are correlated in athletes. *Frontiers in Psychology*, *10*, 576. https://doi.org/10.3389/fpsyg.2019.00576

Mela, C. F., & Kopalle, P. K. (2002). The impact of collinearity on regression analysis: The asymmetric effect of negative and positive correlations. *Applied Economics*, *34*(6), 667–677. https://doi.org/10.1080/00036840110058482

Mesagno, C., & Beckmann, J. (2017). Choking under pressure: Theoretical models and interventions. *Current Opinion in Psychology*, *16*, 170–175. https://doi.org/10.1016/j.copsyc.2017.05.015

Morrison, D. G., & Kalwani, M. U. (1993). The best NFL field goal kickers: Are they lucky or good? *CHANCE*, *6*(3), 30–37. https://doi.org/10.1080/09332480.1993.10542375

Retrieved from https://www.nytimes.com/2007/10/28/sports/playmagazine/28lewis.html

Retrieved from https://www.nytimes.com/2018/09/17/sports/with-nfl-kickers-you-get-what-you-pay-for.html

Toma, M. (2017). Missed shots at the free-throw line: Analyzing the determinants of choking under pressure. *Journal of Sports Economics*, *18*(6), 539–559. https://doi.org/10.1177/1527002515593779

Vickers, J. N. (2007). *Perception, cognition and decision training: The quiet eye in action*. Human Kinetics.

View publication stats